HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 11 Name: _____ Marks: / 101

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The table below shows the distribution of the numbers of keys owned by a group of housewives.

Number of keys	3	4	5	6	7	8
Number of housewives	10	9	4	3	4	k

If a housewife is randomly selected from the group, then the probability that she owns more than 6 keys is $\frac{5}{18}$.

(a) Find k.

(b) Write down the mean, the mode and the median of the distribution.

(5 marks) (DSE-2024 PI Q09)

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2. It is given that $g(x)$ is partly constant and partly varies as x. Suppose that $g(-3) = -21$ and $g(7) = 9$.

(a) Find $g(x)$. (3 marks)

(b) Let $h(x) = xg(x) + k$, where k is a real constant. If all the roots of the equation $h(x) = 0$ are real numbers, find the range of values of k. (3 marks)

(6 marks) (DSE-2024 PI Q10)

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3. The stem-and-leaf diagram below shows the distribution of the numbers of hours spent on reading journals in a month by a group of researchers.

Stem (tens)	Leaf (units)
2	0 0 1 a a a 8 8 9 9
3	0 0 2 3 4 4 7 9
4	0 b

The mean of the distribution is 30 .

(a) Find a and b . (3 marks)

(b) Write down the least possible range of the distribution. (1 mark)

(c) Find the greatest possible inter-quartile range of the distribution. (3 marks)

(7 marks) (DSE-2024 PI Q11)

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4. Denote the origin by O.

(a) A and B are points lying on the positive x-axis such that the x-coordinate of A is greater than the x-coordinate of B. A vertical line which passes through B cuts the straight line $y = mx$ at the point C such that $AB = BC$, where m is a positive constant. Let D be a point such that $ABCD$ is a square. Express the slope of OD in terms of m. (3 marks)

(b) The coordinates of the points M and N are $(6, 5)$ and $(10, 0)$ respectively. Let P and Q be points lying on OM and MN respectively while R and S be points lying on the x-axis. If the quadrilateral $PQRS$ is a square, find the x-coordinate of P. (4 marks)

(7 marks) (DSE-2024 PI Q12)

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5. The base of a solid right pyramid is a square of side 64 cm. The height of the pyramid is 24 cm. The pyramid is divided into a frustum X and a pyramid Y by a plane which is parallel to its base. It is given that the height of Y is 18 cm.

(a) Find the volume of X. (3 marks)

(b) The base of another solid right pyramid is a square. This pyramid is divided into a frustum Z and a pyramid by a plane which is parallel to its base. The height and the total surface area of Z are 3 cm and 960 cm^2 respectively. Are X and Z similar? Explain your answer. (4 marks)

(7 marks) (DSE-2024 PI Q13)

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6. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

(a) Write down the value of p. (1 mark)

(b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

(i) Find q and r.

(ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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7. It is given that $\log_9 y$ is a linear function of $\log_3 x$. Denote the graph of the linear function by L. The slope of L is 4 and L passes through the point $(5, 22)$. Express y in terms of x. (3 marks) (DSE-2024 PI Q15)

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8. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100 ? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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9. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

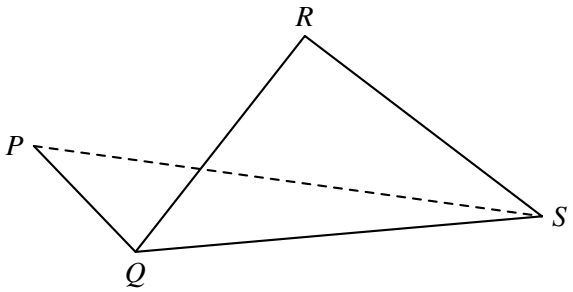


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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10. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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11. Let m be a constant. Solve the equation $x^2 - 3x = (m-1)^2 - 3(m-1)$.

A. $x = m - 1$ or $x = m - 4$

B. $x = m - 1$ or $x = 4 - m$

C. $x = 1 - m$ or $x = m - 4$

D. $x = 1 - m$ or $x = 4 - m$

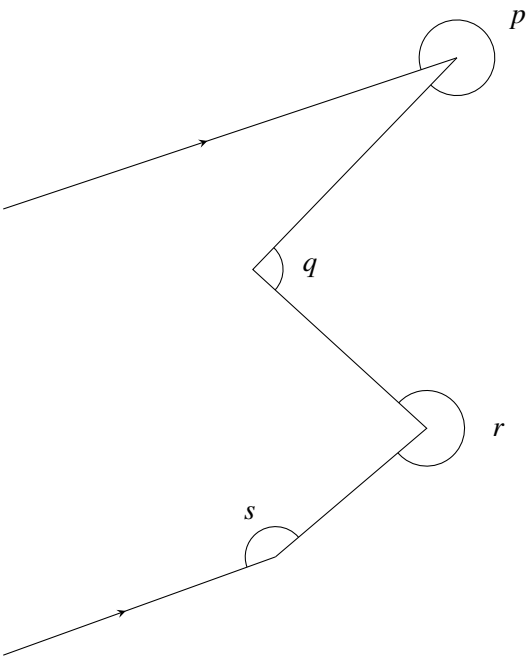
(2 marks) (DSE-2024 PII Q07)

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12. According to the figure, which of the following must be true?



A. $p+q-r=90^\circ$

B. $p-r+s=180^\circ$

C. $p+q-r+s=270^\circ$

D. $p+q+r-s=540^\circ$

(2 marks) (DSE-2024 PII Q19)

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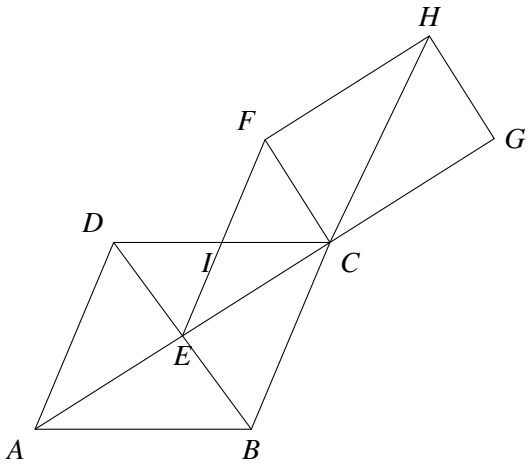
Answers written in the margins will not be marked.

13. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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14. The coordinates of the points A and B are $(-3, 1)$ and $(-7, -5)$ respectively. If P is a point lying on the straight line $x - y + 13 = 0$ such that $AP = PB$, then the y-coordinate of P is

A. -11.

B. -2.

C. 2.

D. 11.

(2 marks) (DSE-2024 PII Q25)

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15. Two numbers are randomly drawn at the same time from six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is not less than 12.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{7}{15}$

D. $\frac{8}{15}$

(2 marks) (DSE-2024 PII Q28)

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16. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

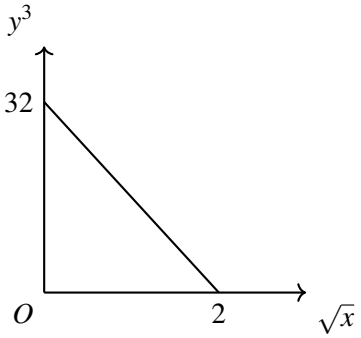
(2 marks) (DSE-2024 PII Q32)

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17. The graph in the figure shows the linear relation between y^3 and $\sqrt{x}$. If $x = 36$, then $y =$



A. -64.

B. -16.

C. -8.

D. -4.

(2 marks) (DSE-2024 PII Q34)

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18. Let $z=(a-5)i+\dfrac{(a+2)i}{2+i}$. If a and z are real numbers, then $a-z=$

A. 2.

B. 3.

C. 4.

D. 5.

(2 marks) (DSE-2024 PII Q35)

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19. The sum of the first n terms of a sequence is $n(2n+3)$. Which of the following are true?

I. 14 is a term of the sequence.

II. The nth term of the sequence is $4n+1$.

III. The sequence is an arithmetic sequence.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

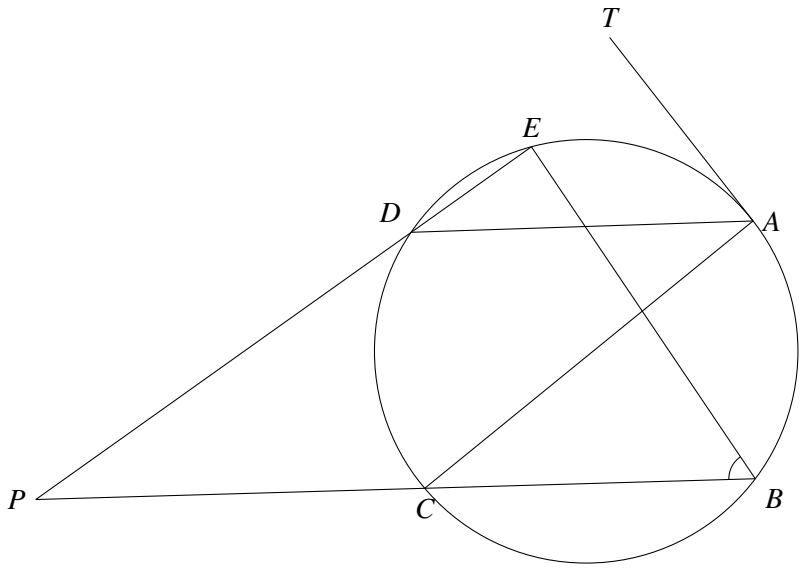
(2 marks) (DSE-2024 PII Q36)

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20. In the figure, TA is the tangent to the circle $ABCDE$ at the point A. BC produced and ED produced meet at the point P. If $\angle ACB = 43°$, $\angle DAT = 55°$ and $\angle CPD = 29°$, then $\angle CBE =$



A. 64°.

B. 69°.

C. 72°.

D. 78°.

(2 marks) (DSE-2024 PII Q38)

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21. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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22. A queue is formed by 2 managers and 7 officers. If no managers are next to each other, how many different queues can be formed?

A. 80 640

B. 141 120

C. 282 240

D. 362 880

(2 marks) (DSE-2024 PII Q42)

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23. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024

B. 0.188

C. 0.812

D. 0.976

(2 marks) (DSE-2024 PII Q43)

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24. In an examination, the scores (in marks) of the candidates are as follows:

39 10 13 16 17 19 25 26 28 30 30 32

Which of the following is/are true?

I. The median of the examination scores of the candidates is 22 marks.

II. The standard score of each candidate in the examination is lower than 2.

III. The standard deviation of the examination scores of the candidates exceeds 8 marks.

A. I only

B. II only

C. I and III only

D. II and III only

(2 marks) (DSE-2024 PII Q44)

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25. If the variance of the seven numbers x_1, x_2, x_3, x_4, x_5, x_6 and x_7 is 16, then the standard deviation of the seven numbers $9x_1 - 5$, $9x_2 - 5$, $9x_3 - 5$, $9x_4 - 5$, $9x_5 - 5$, $9x_6 - 5$ and $9x_7 - 5$ is

A. 31.

B. 36.

C. 139.

D. 144.

(2 marks) (DSE-2024 PII Q45)

I can finish _____ questions

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